

Bank Stocks and Multifactor Models

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Abstract

Multifactor models have been developed over time to price nonfinancial stocks. This paper investigates their ability to price widely-perceived anomalous bank stocks. Based on standard cross-sectional regression tests, we find that, as factors are added to multifactor models, their ability to price bank stocks substantially improves. Even so, a recently proposed two-factor model normally outperforms other multifactor models with relatively high goodness-of-fit and statistical significance. Further graphs of average mispricing errors confirm these findings. We conclude that traditional multifactor models well explain the cross section of average bank stock returns.

Keywords: bank stocks, cross-sectional tests, mispricing error, multifactor models

JEL: G12, C20

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1 Introduction

The now famous Capital Asset Pricing Model (CAPM) of Treynor (1961, 1962), Sharpe (1964), Lintner (1965), and Mossin (1966) is a general equilibrium model that applies to all assets. Recognizing this general market perspective, early empirical tests of the CAPM employed all U.S. stocks. Tests by Black, Jensen, and Scholes (1972), Fama and MacBeth (1973), and others yielded mixed results with a weaker and flatter relationship between beta and stock returns than predicted by theory. In a later series of papers, Fama and French (1992, 1993, 1995, 1996) documented compelling evidence that beta was not significantly related to stock returns, which led them to conclude that the CAPM was dead. In its place, they proposed a three-factor model with market, size, and value factors. The latter two factors were computed from firm characteristics based on market capitalization and the book-to-market equity (BM) ratio, respectively. Since the book values of equity for financial firms are very low compared to nonfinancial firms, they excluded the former from their analyses.

Focusing on nonfinancial firms, researchers expanded the three-factor model over time to include additional factors. For example, Carhart (1997) and later Fama and French (2018) added a momentum factor. Hou, Xue, and Zhang (2015) and Fama and French (2015) incorporated profitability and capital investment factors. Stambaugh and Yuan (2017) constructed management and performance factors. These and other extensions of the three-factor model have sought to explain anomalous stock portfolios. A large literature on stock return anomalies has evolved in tandem with the continued development of multifactor models designed to explain them.¹

While the size of banking institutions was historically constrained prior to 1980, deregulation of interest rates on deposits and interstate banking has enabled U.S. commercial

¹See Fama and French (1998), Cochrane (2011), Chen and Zimmerman (2022), Jensen, Kelly, and Pedersen (2023), and citations therein.

banks to rapidly expand from approximately \$1.4 trillion in total assets in 1980 to over \$24 trillion by 2025.² Relaxing financial services restrictions, the Financial Modernization Act of 1999 enabled banks to own insurance, securities, and other financial firms via the creation of financial holding companies. Also, bank holding companies have expanded their activities into securities and other financial services. As these structural changes occurred over the past 45 years, a wave of mergers and acquisitions increased concentration in the industry, the number of publicly traded banks, and their stock market capitalizations.³ Today, banking institutions are diversified financial service organizations with greater scale and scope economies than in the past.

Relevant to the present study, after more than three decades of intensive development of nonfinancial models, no studies empirically test the extent to which these more advanced models can explain the cross section of bank stock returns. As nonfinancial models have matured over time to include profitability, capital investment, momentum, and other factors, and major structural changes have transformed banking organizations, a natural question is: Are bank stocks anomalous to more advanced multifactor models for nonfinancial firms? That is, are banks so different from nonfinancial firms that they cannot be effectively priced by traditional models? Given their growing importance in the U.S. economy and financial markets, a gap exists in the literature to the extent bank stocks are excluded from conventional asset pricing tests of stock returns.

This paper contributes to the asset pricing literature by investigating whether nonfinancial multifactor models can explain bank stock returns. Using standard Fama and MacBeth (1973) cross-sectional regression analyses, we comparatively study the CAPM plus five well-known multifactor models that have evolved over time. Also, we include the Mean Variance Pricing (MVP) model of Kolari, Huang, Liu, and Liao (2025b). This recent model contains only two factors: (1) the cross-sectional excess market return related to beta risk, and (2) the cross-sectional standard deviation of returns (i.e., market return dispersion) associated with so-called zeta risk.

²FRED database, Federal Reserve Bank of St. Louis.

³The number of FDIC-insured commercial banks decreased from 8,315 in 2000 to 3,928 in 2024. According to Wikipedia, financial services accounted for approximately 15 percent of the S&P 500 equal weight index in 2024.

Using size- and beta-sorted portfolios and out-of-sample cross-sectional regression tests, we find that market factor loadings (i.e., beta estimates) in the CAPM are significantly priced in the cross section of average bank stock returns but contrary to theory have negative signs. Particularly in the case of size-sorted portfolios, as size, value, profit, capital investment, and momentum factors are added to multifactor models, their ability to price bank stocks markedly improves. Also, relative to other multifactor models, we find that the MVP model does a superior job of pricing bank stocks. In size-sorted bank stock portfolios, it achieves almost perfect out-of-sample goodness of fit (viz., R^2 of 99 percent) and has highly significant factor loadings associated with zeta risk (viz., t -statistic exceeding 4). Based on other test asset portfolios sorted on market risks, while leading multifactor models again perform increasingly better as factors are added, the MVP model outperforms them with respect to goodness-of-fit and zeta risk factor loadings' significance. Additionally, augmenting models with interest rate factors, such as term and credit risk spreads, yields mixed findings that improves performance for size-sorted portfolios but not other portfolios sorted on different market risks.

Lastly, graphical analyses of out-of-sample average predicted (fitted) excess returns and actual excess returns show that average mispricing errors decline as factors are added to multifactor models. However, mispricing errors are normally smaller for the two-factor MVP model compared to other models, especially in the case of size-sorted portfolios. To reconcile these findings, discussion is provided in the context of total market dispersion as an asset pricing factor. In this regard, KLH have contended that long/short factors in multifactor models are rough measures of the cross-sectional distribution of returns of stocks in the market. As such, they are subsumed within total market dispersion.

In view of these empirical findings, we conclude that, as factors are added to multifactor models, their ability to price bank stocks improves. Even so, the parsimonious two-factor MVP model outperforms other multifactor models in out-of-sample cross-sectional regression tests. We infer that bank stocks are less anomalous to asset pricing factors used in nonfinancial studies than previously believed.

2 Literature Review

Due to their different firm characteristics, a separate branch of asset pricing literature exists on the valuation of bank stocks. Early studies identified a term structure factor in bank stock returns. For example, Flannery and James (1984) found that interest rate changes affected the stock returns of banks and savings and loan institutions through differences in their asset and liability maturities. Other studies have confirmed this interest rate sensitivity, including Stone (1974), Yourougou (1990), Bae (1990), Madura and Zarruk (1995), and Oertmann, Rendu, and Zimmermann (2000). More generally, Cooper, Jackson, and Patterson (2003) showed that bank stock returns are related to a variety of bank-specific financial ratios, including earnings per share, loans/total assets, loan-loss reserves/total loans, non-interest income/net income, book value of equity/total assets, etc. One-way decile sorts on quarterly percent changes in these variables were significantly associated with bank stock returns. Monthly cross-sectional regressions of stock returns on lagged ratio variables confirmed these significant relationships.

Consistent with the influence of financial condition on bank stocks, Fahlenbrach, Prilmeier, and Stulz (2012) observed that banks with poor stock market performance during the 1998 Russian debt default crisis later underperformed in the 2008-2009 Great Financial Crisis (GFC). These banks tended to have relatively weak fundamental variables, such as funding risk, financial leverage, and asset growth. Also, Adrian and Brunnermeier (2016) found that the systemic risk of financial firms was related to bank-specific variables, including financial leverage, maturity mismatches in assets and liabilities, market capitalization, and market-to-book equity ratios.⁴

In a study of common risk factors, Viale, Kolari, and Fraser (2009) tested alternative asset pricing models using 25 bank stock portfolios formed from the intersection of size and book-to-market quintiles. They compared the CAPM, Fama and French three-factor model, and variants of Merton's (1973) ICAPM (e.g., innovations in dividend yield, interest rates, long- and short-term bond spreads, and default bond spreads). Using monthly stock

⁴For further work on how bank debt and equity values affect the systemic risk of banks, see Acharya, Engle, and Pierret (2014).

returns from 1986 to 2003, cross-sectional regression tests revealed that Fama-French size and value factor loadings as well as default risk loadings were not significant; however, market factor and term factor loadings were statistically significant.

Another study by Bessler and Kurmann (2014) examined the relationship between bank stock returns and various vector autoregression (VAR) innovations in long-term bond rates, bond credit risk spreads, real estate investment trust (REIT) returns, return on a trade-weighted exchange-rate basket of currencies, changes in overnight interbank rates (i.e., bank liquidity risk), and measures of investors' risk aversion.⁵ Time-series regression model tests for the sample period from 1990 to 2011 showed that the market factor was the most significant variable in European and U.S. bank stock portfolios. With the exception of long-term bond rates and exchange rates, other factor innovations were significant also; however, subperiod analyses documented shifting risk exposures of banks over time. Additionally, they found that small and large bank stocks exhibited differences in their risk exposures.

Gandhi and Lustig (2015) used size-sorted portfolios of bank stocks to conduct asset pricing tests. They identified a significant second principal component corresponding to a new size factor. This factor was different from the market capitalization factor of Fama and French for nonfinancial firms. They conjectured that, in bad economic times, small banks are more negatively affected than large banks, and their recoveries from post-economic shocks are slower despite being less levered than large banks. Thus, they inferred that the component size factor measures bank-specific tail risk orthogonal to nonfinancial firm risk factors.

Related to these findings, Bouwman, Kim, and Shin (2023) found that, during bad economic times, high-capital banks had higher risk-adjusted stock returns than low-capital banks. Since bank capital is measured as common book equity to total assets, their evidence supports the existence of a value factor (i.e., proxied by the ratio of book equity to market equity) in bank stock returns. The significance of bank capital in bad times but not other times suggests that this factor is conditional on economic conditions. They

⁵For interested readers, citations are provided for studies on bank stock returns and exchange rate risk as well as real estate risk.

inferred that capital is beneficial to shareholders in terms of enabling banks to withstand business cycle downturns. Earlier evidence by Berger and Bouwman (2013) corroborates that medium and large banks' performance is improved by capital during banking crises. For small banks, regardless of economic conditions, capital boosted survival probabilities and market share.

In sum, previous bank stock studies found that interest rates and bank financial condition are potentially significant pricing factors. Mixed findings were reported for size and value factors with more recent studies detecting significance for these factors. We extend this literature by conducting standard cross-sectional regression tests of different multifactor models in addition to interest rate factors. By testing multifactor models developed to price nonfinancial stocks, we contribute new evidence on bank asset pricing.

3 Methodology

3.1 Data

Daily returns for bank stocks are gathered from the Center for Research in Security Prices (CRSP) database. The Federal Reserve Bank of New York links regulatory identification numbers (RSSD ID) from the National Information Center (NIC) to the permanent company number (PERMCO) in CRSP. There are 1,471 RSSD-PERMCO links from June 30, 1986 to September 29, 2023. For each sample bank, we collect returns from January 1986 to December 2023.

For test assets, we construct the following bank stock portfolios: 25 size-sorted, 25 beta-sorted, 25 zeta-sorted, and 25 beta-zeta sorted. Monthly market capitalization (i.e., number of shares outstanding times stock price) is used to rank bank stocks and form 25 size portfolios. Daily equal-weighted average returns from January 1986 to December 2023 are computed for each portfolio. To obtain 25 beta portfolios, we implement the following steps: (1) based on one year of daily returns starting in 1986, market beta is estimated using the CAPM market model for each bank stock; (2) 25 beta-sorted portfolios are formed; (3) in the next month ahead of the one year estimation period starting in January

1987, equal-weighted daily returns are computed for each portfolio; and (4) this process is rolled forward one month at a time and repeated to construct an average daily return series for each beta-sorted portfolio from January 1987 to December 2023. Based on the MVP model (to be discussed shortly), this process is used to similarly form 25 zeta-sorted and 25 beta-zeta sorted portfolios.

Figures 1 and 2 show the mean daily returns in the sample period for the test asset portfolios. Panel A of Figure 1 shows that mean bank returns and bank size (market capitalization) are inversely related to one another. Smaller banks on the left side of the graph have noticeable higher mean returns that gradually decline as bank size increases on the right side of the graph. The range of mean returns is considerable from 0.19 percent per day for small banks to 0.05 percent per day for big banks. Beta-sorted portfolios in panel B exhibit a much narrower return range from slightly more than 0.04 percent per day to 0.07 percent per day. Also, mean returns are flat for the most part as beta increases with the exception of high beta portfolios with somewhat higher returns. Likewise, zeta-sorted portfolios based on the MVP model in panel A of Figure 2 have a flat pattern of mean returns with the exception of low zeta portfolios with somewhat lower returns. Finally, in panel B using beta-zeta sorted portfolios, which are formed by creating beta quintiles and then zeta quintiles within each beta quintile, mean returns are flat with the exceptions of low-beta/low-zeta portfolios with lower returns and high-beta/high-zeta portfolios with higher returns.

The value-weighted CRSP index minus the Treasury bill rate proxies the market factor. Size, value, momentum, profitability, and capital investment factors are downloaded from Kenneth French’s online database. The market dispersion factor for the MVP model is the cross-sectional standard deviation (or second moment) for all U.S. stocks on each day t computed as follows:

$$\sigma_{mt} = \sqrt{\frac{n}{n-1} \sum_{i=1}^n w_{it-1} (R_{it} - R_{mt})^2}, \quad (1)$$

where n is the total number of stocks; w_{it-1} is the previous day’s market value weight for

the i th stock (i.e., market capitalization of the stock divided by the market capitalization of all n stocks); R_{it} is the return of the i th stock on day t ; and R_{mt} is value-weighted average market return of all available stocks in the CRSP database on day t . Also, in line with previous bank stock studies, we include two state variables associated with interest rate and credit risks: (1) *TERM* is the yield spread between 10-year Treasury bond and 3-month Treasury bill yields; and (2) *DEF* is the credit risk spread between corporate BAA and 10-year Treasury bond yields. Both series are downloaded from the FRED online database at the Federal Reserve Bank of St. Louis.

Descriptive statistics for asset pricing factors in different models are summarized in Table 1. Panel A (B) contains distribution statistics (correlation coefficients). In panel A, note that the mean daily market return dispersion factor σ_m is relatively large at 1.90 percent per day compared to other factor returns ranging from a mean daily return of 0.002 percent for the size factor *SMB* to 0.04 percent for the market factor $R_m - R_f$. Hence, the magnitude of cross-sectional market volatility is large compared to other factors on any given day. Panel B reports the correlation coefficients for different factors. The highest correlation is 0.54 between the *HML* and *CMA* factors.⁶ Also, *HML* has a correlation coefficient of -0.32 with *MOM*. The market factor $R_m - R_f$ exhibits a similar level of correlation with the *HML*, *RMW*, and *CMA* factors (viz., estimated coefficients in the range of -0.29 to -0.33). Finally, cross-sectional market volatility σ_m is not correlated with the market factor and other stock factors but has a correlation coefficient of 0.27 with credit risk spread *DEF*.

3.2 Cross-sectional regression analyses

To compare different asset pricing models, we implement the standard two-step Fama and MacBeth (1973) procedure. First, using one year of daily returns from January 1987 to December 1987, each model is estimated via the following time-series regression:

$$R_{it} - R_{ft} = \alpha_i + \sum_{k=1}^K b_{ik} F_{kt} + e_{it}, \quad t = 1, \dots, T, \quad (2)$$

⁶Fama and French (2015) dropped the *HML* factor from their five-factor model including *RMW* and *CMA* factors in all likelihood due to this correlation.

where $R_{it} - R_{ft}$ is the realized excess return on the i th test asset for day t ; R_{ft} is the riskless rate proxied by the Treasury bill rate; α_i is the intercept term or alpha, F_{kt} are asset pricing factors for $k = 1, \dots, K$ factors, b_{ik} are corresponding K factor risk loadings for the i th portfolio; $i = 1, \dots, N$ correspond to the number of portfolios; $t = 1, \dots, T$ is the estimation period; and $e_{it} \sim \text{iid}(0, \sigma_i^2)$.

Second, an out-of-sample (one-month-ahead) cross-sectional regression is estimated. Many researchers consider out-of-sample tests to be less prone to data snooping and econometric problems and, therefore, more reliable than in-sample tests.⁷ For each model, the following one-month-ahead cross-sectional regression is run:⁸

$$R_{iT+1} - R_{fT+1} = \lambda_0 + \lambda_1 \hat{b}_{i1} + \lambda_2 \hat{b}_{i2} + \dots + \lambda_K \hat{b}_{iK} + u_{iT+1}, \quad i = 1, \dots, N, \quad (3)$$

where $R_{iT+1} - R_{fT+1}$ is the realized excess return on the i th portfolio in out-of-sample (one-month-ahead) month $T + 1$ January 1988; λ_{i0} is the intercept⁹; $\hat{b}_{ik}, k = 1, \dots, K$ are K factor risk loadings estimated for N portfolios in the estimation period $t = 1, \dots, T$; λ_k are corresponding estimated market prices of factor risk loadings¹⁰; and u_{iT+1} are residuals that are mean zero and independent of the explanatory variables.

The above two-step procedure is rolled forward one month at a time and repeated until December 2023. A monthly series of λ_k parameters is generated. For the analysis period from January 1988 to December 2023, average estimates for λ_k and associated t -statistics are computed. Additionally, for each portfolio in the analysis period, we compute monthly time series of realized and predicted returns that are averaged. To estimate the R^2 for each model, in line with Jagannathan and Wang (1996) and Lettau and Ludvigson (2001, footnote 17, p. 1254), average realized returns are regressed on average predicted returns. Subsequently, these average realized and predicted returns are graphically plotted to

⁷See Simin (2008), Ferson, Nallareddy, and Xie (2013), Barillas and Shanken (2018), Fama and French (2018), and others.

⁸Following Giglio and Xiu (2021), we use OLS regression analyses. The authors found that OLS estimates were not improved by other econometric approaches, including weighted least squares (WLS), generalized least squares (GLS), general method of moments (GMM), and maximum likelihood (ML).

⁹According to Fama and French (2020), the intercept represents the common return to all test assets not captured by the explanatory variables.

¹⁰The market prices of risk represent long-short mimicking portfolios of the factor loadings of stock portfolios. See Ferson, Sarkissian, and Simon (1999), Cochrane (2005), Back, Kapadia, Ostdiek (2013, 2015), Ferson (2019), and others.

provide a visual picture of average mispricing errors for bank stock portfolios.

3.3 Asset pricing models

In our empirical tests, we include the CAPM as well prominent multifactor models developed over time to price nonfinancial stocks. The CAPM market model is estimated using the value-weighted CRSP index minus the Treasury bill rate to proxy the market factor ($R_m - R_f$). The Fama and French (1992, 1993) three-factor model (FF3) includes the market factor plus size factor (small minus large firms' stock returns, or *SMB*) and value factor (high book-to-market equity minus low book-to-market equity firms' stock returns, or *HML*). Carhart's (1997) four-factor model (C4) augments FF3 with a momentum factor (firms with high past return stock returns minus low past stock returns, or *MOM*). Extending FF3, the Fama and French's (2015) five-factor model (FF5) adds a profitability factor (robust operating profitability minus weak operating profitability returns, or *RMW*) and capital investment factor (conservative investment minus aggressive investment returns, or *CMA*). Also, the Fama and French (2018) six-factor model (FF6) augments their FF5 model with a momentum factor.

Additionally, we include a recent model dubbed the Mean Variance Pricing (MVP) model by Kolari, Huang, Liu, and Liao (2025b). This parsimonious two-factor model is comprised of a market factor plus a cross-sectional market return dispersion factor. The latter cross-sectional market volatility has been shown by many studies to be related to macroeconomic variables, including unemployment, economic uncertainty, economic restructuring, business cycles, and financial market volatility.¹¹ Since the MVP model was proposed in a recent working paper and performs well in forthcoming asset pricing tests, we next provide an overview that summarizes its theoretical and empirical specifications.

3.3.1 The ZCAPM and MVP model

Returning to general equilibrium CAPM theory, Kolari, Liu, and Huang (KLH) (2021) derived a special case of Black's (1972) zero-beta CAPM dubbed the ZCAPM. As a starting

¹¹See Pastor and Veronesi (2009), Bansal, Kiku, Shaliastovich, and Yaron (2014), Garcia, Mantilla-Garcia, and Martellini (2014), Angelidis, Sakkas, and Tessaromatis (2015), and citations therein.

point, the authors proved that the width or span of the mean-variance investment parabola of Markowitz (1952, 1959) is determined by the cross-sectional standard deviation of assets' returns in the market. Given this new insight, the mean market return lies approximately on the axis of symmetry of the parabola. Figure 3 provides a graphical depiction of the ZCAPM. Portfolio I^* on the efficient frontier of the mean-variance parabola has zero-beta counterpart portfolio ZI^* . The expected returns of these portfolios can be specified in terms of the expected mean market return (denoted m) and the cross-sectional standard deviation of returns of all assets in the market, or σ_m , as follows:

$$E(R_{I^*}) \approx E(R_m) + \sigma_m \quad (4)$$

$$E(R_{ZI^*}) \approx E(R_m) - \sigma_m. \quad (5)$$

Black's (1972) zero-beta CAPM is:

$$E(R_i) = E(R_{ZM}) + \beta_{iM}[E(R_M) - E(R_{ZM})], \quad (6)$$

where $E(R_M)$ is the expected market portfolio return; $E(R_{ZM})$ is the expected zero-beta portfolio return that is uncorrelated (orthogonal) to the market portfolio; and β_{iM} is beta risk associated with excess expected market portfolio M returns. Roll (1972), Copeland and Weston (1980), and others re-write Black's model in more general form as:

$$E(R_i) = E(R_{ZI}) + \beta_{iI}[E(R_I) - E(R_{ZI})], \quad (7)$$

where I and ZI are any efficient portfolio and its orthogonal zero-beta portfolio counterpart, respectively.

Substituting the expected returns for portfolios I^* and ZI^* in equations (4) and (5) into zero-beta CAPM relation (7), the theoretical ZCAPM can be derived as follows:

$$\begin{aligned} E(R_i) &= E(R_{ZI^*}) + \beta_{iI^*}[E(R_{I^*}) - E(R_{ZI^*})] \\ &= E(R_m) - \sigma_m + \beta_{iI^*}\{[E(R_m) + \sigma_m] - [E(R_m) - \sigma_m]\} \\ &= E(R_m) + (2\beta_{iI^*} - 1)\sigma_m \\ E(R_i) &= E(R_m) + Z_{im}^*\sigma_m, \end{aligned} \quad (8)$$

where $Z_{ia}^* = 2\beta_{iI^*} - 1$ (i.e., zeta risk associated with σ_a). As shown by K LH, given the

existence of a riskless asset, the ZCAPM becomes:

$$E(R_i) - R_f = \beta_{im}[E(R_m) - R_f] + Z_{im}^* \sigma_m, \quad (9)$$

where β_{im} represents beta risk associated with expected market excess returns; and Z_{im}^* is zeta risk related to the cross-sectional market return dispersion.

In their empirical specification of the ZCAPM, the first and second moments of assets' returns in the U.S. stock market – namely, the value-weighted CRSP index return and its value-weighted standard deviation on any given day t – are employed in constructing a market factor (in excess of the Treasury bill rate) and market return dispersion factor, respectively. They proposed the following two-factor model:

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + Z_i D_{it} \sigma_{mt} + u_{it}, \quad t = 1, \dots, T, \quad (10)$$

where Z_i measures zeta risk; D_{it} is a hidden signal variable with dummy values $+1$ and -1 to coincide with positive and negative market return dispersion effects on asset returns at time t , respectively; α_i is the intercept related to mispricing errors; $u_{it} \sim \text{iid } N(0, \sigma_i^2)$; and other notation is as before. To estimate the probability p_i of hidden variable D_{it} , the authors implemented a relatively sophisticated expectation-maximization (EM) algorithm.¹² In turn, the zeta risk coefficient Z_i^* in ZCAPM relation (4) is estimated as $(2p_i - 1)Z_i$. Also, they found that inclusion of α_i in the model increased residual errors and so dropped this intercept parameter (hereafter the EM ZCAPM). In a series of articles and books¹³, based on different test assets and time periods, KLH and co-authors showed that the EM ZCAPM outperformed well-known multifactor models in standard cross-sectional regression tests in terms of goodness-of-fit and statistical significance of zeta risk loadings.

In a recent working paper, Kolari, Huang, Liu, and Liao (2025b) proposed a simpler approach to estimate the ZCAPM. Unlike the more complex EM ZCAPM, a Mean Variance Pricing (MVP) model is specified using standard regression methods. In this respect, previous studies by Jiang (2010), Demirer and Jategaonkar (2013), and Garcia,

¹²See Dempster, Laird, and Rubin (1977), Jones and McLachlan (1990), McLachlan and Peel (2000), McLachlan and Krishnan (2008), and others.

¹³See Liu (2013), Kolari, Huang, Butt, and Liao (2022), Kolari, Huang, Liu, and Liao (2022), Kolari and Pynnonen (2023), Kolari, Liu, and Pynnonen (2024), Kolari, Huang, Liu, and Liao (2025a), Kolari, Huang, Liu, and Liao (2025b), and Kolari, Liu, Huang, and Liao (2026).

Mantilla-Garcia, and Martellini (2014) augmented the market factor with a market return dispersion factor in a multifactor regression model but obtained mixed findings and generally weak overall model performance. The MVP model modifies this two-factor regression approach by setting the intercept to zero:

$$R_{it} - R_{ft} = \beta_i(R_{mt} - R_{ft}) + Z_i\sigma_{mt} + u_{it}, \quad t = 1, \dots, T, \quad (11)$$

where residual error terms u_{it} do not necessarily have mean zero as in ordinary least squares (OLS) estimation. Widely-available regression analysis software can be used to compute standard errors and t -statistics for estimated coefficients in this model.¹⁴ In out-of-sample cross-sectional regression analyses, they found that the MVP model markedly increased goodness-of-fit as measured by R^2 values, and zeta risk loadings associated with market return dispersion were highly significant. Surprisingly, the empirical results were similar to those for the EM ZCAPM.

It is common practice to include an intercept parameter in the empirical estimation of asset pricing models.¹⁵ Unlike these models, both empirical specifications of the ZCAPM – the EM ZCAPM and MVP model – have no intercept. As mentioned above, KLH dropped the intercept parameter in the EM ZCAPM due to increased residual errors. By contrast, Kolari et al. (2025b) dropped the intercept in the MVP model for economic reasons. If market return dispersion is zero, the Markowitz investment parabola would collapse to the axis of symmetry. In this case, the mean market return R_m would equal the riskless rate R_f such that the market factor is zero. When both independent variables (or market factors) in the MVP model equal zero, the excess return for all assets must be zero.¹⁶ Consequently, the MVP model is correctly specified as a regression through the origin.

¹⁴Given a two-factor model, the t -statistic for a regression coefficient is defined as $\hat{\beta}_j/\text{SE}(\hat{\beta}_j)$. Coefficient $\mathbf{c}^T\hat{\boldsymbol{\beta}}$ follows a normal distribution $N(\mathbf{c}^T\boldsymbol{\beta}, \sigma^2\mathbf{c}^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{c})$, where MSE is used to estimate σ^2 , and $\mathbf{c} = (1, 0)$ or $\mathbf{c} = (0, 1)$ can be used to obtain the distribution for β_{im} (or Z_{im}). In the regression model with intercept term, $\text{MSE} = \sum_{i=1}^n (y_i - \hat{y})^2 / (n - 3)$, whereas in the MVP regression model without intercept, $\text{MSE} = \sum_{i=1}^n (y_i - \hat{y})^2 / (n - 2)$ with corresponding $\hat{y}_i$ values.

¹⁵Jensen (1968) proposed that the CAPM could be tested using the intercept, or alpha, to measure mispricing error. Gibbons, Ross, and Shanken (GRS) (1989) later developed a more formal test of the joint equality of alphas equal to zero for different test assets. This widely-used test is employed to evaluate the validity of an asset pricing model.

¹⁶By contrast, in the market model estimation of the CAPM, if the market factor is set to zero (e.g., the CRSP value-weighted return equals the Treasury bill rate), the excess return on the i th asset may well be positive due to missing factors. The intercept parameter (or α_i) in the market model explicitly recognizes this possibility. Multifactor models contain an intercept parameter for these same reason.

Previously cited studies that augmented the market factor with a market return dispersion factor incorrectly included an intercept term in their models. This misspecification was due to not recognizing that market return dispersion determines the width or span of the investment parabola as proven mathematically in Kolari, Liu, and Huang (2021, Chapters 3 and 4).

4 Empirical Results

In this section, we report the cross-sectional regression results as well as graphs of average mispricing errors for different asset pricing models and test asset portfolios.

4.1 Cross-sectional regression results

Table 2 shows the results for out-of-sample (one-month-ahead) Fama-MacBeth cross-sectional regressions. Panel A contains the findings for 25 size-sorted portfolios. In terms of goodness-of-fit, estimated R^2 values range from 43.0 percent for the CAPM to 99.0 percent for the MVP model. The FF6 model achieves an R^2 value of 78.0 percent, which is very good but does not match the near perfect goodness-of-fit of the MVP model. These out-of-sample results suggest that the cross section of bank stock returns is almost completely explained by beta and zeta risk loadings in the MVP model.

Not surprisingly, given the explanatory power of the MVP model, the estimated market price of zeta risk loadings, or $\hat{\lambda}_{\sigma_m}$, indicates a relatively high risk premium equal to 0.49 percent per month.¹⁷ Note that the associated t -statistic is 4.19, which is highly significant at lower than the one percent level. To avoid false discoveries of asset pricing factors, Harvey, Liu, and Zhu (2016) and Chordia, Goyal, and Saretto (2020) have recommended a t -statistic threshold of 3.¹⁸ Next most significant at the five percent level is the market price of momentum loadings, or $\hat{\lambda}_{MOM}$, with t -statistics equal to -2.46 and -2.64 in the

¹⁷We should note that, for both the MVP model with no intercept and EM ZCAPM, the zeta risk coefficient or factor loading is rescaled from a daily basis to a monthly basis via multiplying by 21 (i.e., the number of trading days per month). For further discussion, see Kolari, Liu, and Huang (2021, p. 127).

¹⁸In unreported results, the results for the EM ZCAPM are very similar to those for the MVP model. The R^2 value is 99 percent, market price of zeta risk loadings is 0.37 percent per month, and associated t -statistic is highly significant at 3.25.

C4 and FF6 models, respectively. Additionally, in the FF5 model, the market price of profitability loadings, or $\hat{\lambda}_{RMW}$, is nominally significant at the 10 percent level with a t -statistic of -1.83.

Interestingly, the market price of beta risk loadings, or $\hat{\lambda}_m$, are negative and significant at 1 percent to 10 percent levels in all models. Contrary to CAPM theory, the market factor is negatively priced in the cross section of bank stock returns. This negative relation could be due to supervisory oversight of risk taking by bank regulators. As banks increase their exposure to general market risk, regulatory supervision and costs increase, which tend to lower stock returns. By contrast, in tests of multifactor models using nonfinancial firms, beta risk is normally not significantly priced.

Panel B of Table 2 contains the results for 25 beta-sorted portfolios. Again the MVP model has a noticeably higher R^2 value at 82 percent compared to at most 40 percent for the FF6 model. The CAPM performs poorly with an R^2 value at 13 percent, which is improved by the FF3 model at 37 percent. Nonetheless, the MVP model well outperforms other models in terms of reliably predicting monthly bank stock returns on an out-of-sample basis.

The only significant market price of risk in panel B of Table 2 is associated with zeta risk loadings, or $\hat{\lambda}_{\sigma_m}$. The estimated coefficient is -0.21 percent per month with t -statistic equal to -2.27, which is significant at the 5 percent level. In panel A, we found that beta loadings were negatively priced among size-sorted portfolios. Thus, we infer that, given beta risk, average stock returns decrease as risk increases in zeta-sorted portfolios also. At higher risk levels, regulatory pressures increase that tend to lower bank stock returns.

Panel C of Table 2 reports the findings for 25 zeta-sorted portfolios. Now the market price of beta risk loadings, or $\hat{\lambda}_m$, is positive and significant in the C4, FF6, and MVP models. Hence, conditional on zeta risk levels, beta loadings are positively priced in line with CAPM theory. In the MVP model, the market price of zeta risk loadings, or $\hat{\lambda}_{\sigma_m}$, equals 0.22 percent per month with a t -statistic of 2.50, which is significant at the 5 percent level. The only other factor loadings that are significantly priced (at the 5 percent level) in the other models are size risk loadings, or $\hat{\lambda}_{SMB}$ in the FF3, C4, FF5, and FF6

models. The negative market price of risk indicates that larger banks have higher returns than those of smaller banks (see Figure 1). Likewise, size loadings are negatively priced in panels A and B but are insignificant.

The results in panel D of Table 2 for 25 beta-zeta-sorted portfolios are similar to those for beta-sorted portfolios in panel B. However, now size factor (*SMB*) loadings are significant at the 10 percent level. The MVP model again dominates other models with a noticeably higher R^2 estimate of 74 percent and significant market price of risk of zeta risk loadings, or $\hat{\lambda}_{\sigma_m}$, at close to the 5 percent level.

Lastly, we augment all models with the interest rate factors TERM and DEF (viz., the yield spreads between long- and short-term Treasury securities as well as BAA corporate bonds and Treasury bonds, respectively). In unreported results, for 25 size-sorted portfolios, both TERM and DEFAULT loadings are significant at the nominal 10 percent and 5 percent levels in most models. In test asset portfolios, R^2 values increased to some extent for all models, with the exception of the MVP model in which R^2 estimates decreased (e.g., from 99 percent to 88 percent).¹⁹ For the beta-, zeta-, and beta-zeta-sorted portfolios, the interest rate factors' loadings are predominantly insignificant and do not improve R^2 values. We infer that, with the exception of the MVP model, interest rate factors can enhance the ability of multifactor models to price size-sorted portfolios but not other portfolios sorted on different market risks. It appears that nonfinancial factors are sufficient in most cases for pricing bank stocks.

In sum, multifactor models exhibit good pricing ability among anomalous bank stock portfolios that improves as factors are added to the models. However, based on only two market factors, the MVP model normally outperforms other multifactor models.

4.2 Graphical analyses of average mispricing errors

Cochrane (1996), Lettau and Ludvigson (2001), and others have argued that estimation of mispricing errors is recommended to evaluate asset pricing models. These errors can

¹⁹As observed earlier in Table 1, cross-sectional market volatility is correlated with bond credit risk spreads to some extent, which accounts in part for this result. In footnote 11, we cited studies that link market return dispersion with business cycles and financial market volatility. These macroeconomic shocks would affect bond credit spreads.

be computed as mean predicted returns minus mean realized (actual) returns for test asset portfolios. Based on cross-sectional regressions, we utilize out-of-sample (one-month-ahead) predicted excess returns for our test asset portfolios for different asset pricing models. The analysis is rolled forward one month at a time to generate a monthly time series of predicted (and realized) excess returns for each portfolio. Mean values of predicted (X-axis) and realized (Y-axis) returns are graphed to assess average mispricing errors. If all portfolios lie on a 45 degree line from the origin, there are no mispricing errors. As dispersion around the line of portfolios increases, mispricing errors increase.

For the 25 size-sorted portfolios, Figures 4 to 6 illustrate the results for different models under study. In panels A and B of Figure 4, we see that the average mispricing errors for the CAPM and FF3 model, respectively, are similar and not large in magnitude. In Figure 5, average mispricing errors for the C4 (panel A) and FF5 (panel B) models decrease somewhat with some exceptions for a few portfolios relatively distant from the 45 degree line. Finally, comparing the FF6 (panel A) and MVP (panel B) models in Figure 6, average mispricing errors are markedly smaller for the MVP model. Indeed, mispricing errors are near zero for the MVP model.

To conserve space, for other test asset portfolios, we only report results for the FF6 and MVP models. Graphs for other models are available upon request from the authors. Figure 7 shows the graphs for the 25 beta-sorted portfolios. Again the MVP model has visibly lower average mispricing errors than the FF6 model. Using 25 zeta-sorted portfolios in Figure 8, the FF6 and MVP models have similar mispricing, albeit fairly strong results in both cases. Finally, for 25 beta-zeta-sorted portfolios, Figure 9 shows that the MVP model outperforms the FF6 model in terms of average mispricing errors, but both models perform well.

In sum, our findings suggest that bank stocks are not anomalous to more advanced multifactor models developed in nonfinancial asset pricing studies as average mispricing errors are not large. Nonetheless, in most test asset portfolios, the two-factor MVP model has lower mispricing errors than other models with greater numbers of factors. Related to this finding, KLH posited that long/short factors in multifactor models are

subsumed within the total market dispersion factor of the ZCAPM. They interpreted the long/short factors (e.g., small stocks with higher average returns and big stocks with lower average returns) as rough measures of market return dispersion or slices within the cross-sectional distribution of stock returns. As long/short factors are added to multifactor models, they more closely approximate the total market dispersion. This perspective explains the increasing ability of multifactor models to explain bank stocks in our analyses. Nonetheless, consistent with our MVP model results, the total market dispersion factor supplants all of the long/short factors and outperforms them. Of course, as more factors are added to multifactor models, multicollinearity can increasingly bias factor loadings. Also, it becomes difficult to find new factors that are orthogonal (or uncorrelated) with respect to other factors. The MVP model resolves these econometric issues. Particularly relevant to the purpose of the present study, market return dispersion affects all stocks, both financial and nonfinancial. Similarly, as multifactor models utilize more factors and thereby capture larger amounts of total market dispersion, they are better able to price both financial and nonfinancial stocks.

4.3 Robustness checks

A participant at the Western Economic Association conference in Denver, Colorado in June 2026 suggested that the multifactor models be estimated using factors developed from bank stocks rather than nonfinancial firms. The main argument was that the factors were developed by Fama and French for sample nonfinancial firms in their studies that excluded financial firms. To compare the MVP model to other well-known models, the factors should be developed for bank stocks and exclude nonfinancial stocks.

For this purpose, we constructed cross section factors from bank stock portfolios. To begin, for each day t within the one-year estimation window from January 1987 to December 1987, we estimate the following in-sample cross-sectional regression:

$$\begin{aligned}
R_{it} - R_{ft} = & \alpha_t + \hat{\lambda}_{mt}\hat{\beta}_{im} + \lambda_{CSMBt}\hat{\beta}_{iSMB} + \lambda_{CHMLt}\hat{\beta}_{iHML} + \lambda_{CMOMt}\hat{\beta}_{iMOM} \\
& + \lambda_{CRMWt}\hat{\beta}_{iRMW} + \lambda_{CCMAt}\hat{\beta}_{iCMA} + \varepsilon_{it}, \quad i = 1, \dots, 30.
\end{aligned} \tag{12}$$

where $\lambda_{kt} = 1, \dots, T$ is the estimated market price of of the k th factor loading on day t , and other notation is as before. Following Fama and French (2020), λ_{kt} estimates represent cross section factors, which we denote by adding the letter C to each factor (e.g., $CSMB$ is the cross section factor for the size factor SMB). As Ferson (2019, p. 223) has observed, cross section factors are long/short portfolios that are long the firm or stock characteristic and short the firm or stock characteristic. Subsequently, we roll forward one year and estimate the time-series regression model from January 1988 to December 1988. The above cross-sectional regression is estimated for all days t in this one year period. We continue to roll forward one year at a time until the end of the sample period in December 2023. This process generates a daily series of λ_{kt} for the cross section factor for all days from January 1987 to December 2023. Like Fama and French (2020), we substitute these cross section factors for their original factor counterparts and re-run equations (2) and (3).

In unreported results available from the authors, the correlation coefficients of the cross section factors with their original factor counterparts were very low ranging from -0.04 to -0.10. Also, the inter-correlations of the cross section factors with each other are low (i.e., the maximum correlation is only 0.17 between $CHML$ and $CCMA$). Thus, the cross section factors constructed from bank stock returns are very different from the original factors developed from nonfinancial stock returns.

The cross-sectional regression results for FF3, C4, FF5, and FF6 models using cross section factors developed from bank stocks are shown in Table 3. In general, the results are mixed, with some models improving relative to the original models in Table 2 and other models not performing as well. Unlike the Table 2 results using nonfinancial factors, the cross section factor for size, or $CSMB$, has significant factor loadings in Panel B with the 25 beta-sorted test assets portfolios. As before, market factor loadings were significantly negatively priced in Panel A with the 25 size-sorted portfolios. The only other significant loadings are for cross section profitability loadings in Panel C with the 25 zeta-sorted portfolios for the FF6 model. Overall, the models augmented with cross section bank factors yield results that are comparable to those for the original models. In

sum, like the nonfinancial factors of Fama and French, cross section factors developed from bank stock returns for these nonfinancial factors are normally not significant in terms of explaining out-of-sample bank stock returns.

5 Conclusion

Multifactor models have added numerous factors over time in an effort to price the growing list of stock return anomalies in the finance literature. Typically, financial stocks are excluded from mainstream asset pricing tests, which are considered to be anomalous due to their unique financial characteristics. Here we test the ability of traditional multifactor models to price bank stocks, including the CAPM, Fama and French’s three-, five-, and six-factor models, Carhart’s four-factor model, and the recent two-factor Mean Variance Pricing (MVP) model by Kolari, Huang, Liu, and Liao. In the sample period January 1986 to December 2023, bank stocks are sorted into different test asset portfolios based on their firm and market characteristics.

Our empirical tests indicate that, as factors are added to nonfinancial multifactor models, their ability to price bank stocks increases. Indeed, based on out-of-sample cross-sectional regression tests of size-sorted bank stock portfolios, the Fama and French six-factor model attained fairly high explanatory power with R^2 equal to 78 percent. Contrary to this trend, a near perfect 99 percent R^2 value was achieved by the two-factor MVP model for these test assets. Using other test asset portfolios, multifactor models again exhibited fairly good explanatory power with improved pricing performance as factors were added to them. However, the parsimonious MVP model outperformed other multifactor models for the most part. The most significant factor loadings across different models were obtained in the MVP model, with a t -statistic as high as 4.2 for zeta risk loadings associated with the market return dispersion factor. In other multifactor models, factor loadings for momentum, profitability, and size were significant at times across test asset portfolios and models. Furthermore, mixed findings for factor loadings related to term and default spreads were obtained. These interest rate factors were significant for

size-sorted portfolios but not generally for other portfolios sorted on various market risks. Lastly, robustness checks using cross section factors developed from bank stock returns did not generally improve the performance of multifactor models that include long/short firm and stock characteristic factors.

Kolari, Liu, and Huang (2021) interpreted long/short factors in multifactor models as rough measures of total market dispersion. Long (short) portfolios have high (low) average returns that represent slices within the cross-sectional return distribution of stocks in the market. As factors are added to multifactor models, they increasingly approximate total market dispersion. Numerous multifactors are subsumed within total market dispersion. Our MVP model results are consistent with this interpretation. Relevant to the present study, all stocks are impacted by total market dispersion, both financial and nonfinancial stocks. Hence, as factors have been added to multifactor models, their ability to price bank stocks has improved.

Based on our empirical evidence, we conclude that, as factors have been added to multifactor models over time to better approximate total market dispersion, their ability to price anomalous banks stocks has substantially improved. In this respect, a more parsimonious approach is to employ the economical two-factor MVP model, which normally outperformed other models in terms of explaining the cross section of average bank stock returns. An important implication of our findings is that asset pricing models should be tested using both financial and nonfinancial stocks. Further tests of multifactor models with regard to financial companies are recommended, including securities firms, insurance companies, and other financial institutions.

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Table 1: Descriptive statistics for asset pricing factors

This table summarizes the descriptive statistics for the asset pricing factors in the sample period January 1986 to December 2023. Daily factor returns (in percent) corresponding to different asset pricing models (see Section 3.3 in the text) are denoted as follows: $R_m - R_f$ (value-weighted CRSP return minus Treasury bill rate), σ_m (market return dispersion), SMB (size), HML (value), MOM (momentum), RMW (profitability), CMA (capital investment), $TERM$ (Treasury spread between 10-year bonds and 3-month Tbills), DEF (credit risk spread between BAA corporate bonds and 10-year Treasury bonds).

Panel A: Distribution statistics									
Statistic	$R_m - R_f$	σ_m	SMB	HML	MOM	RMW	CMA	$TERM$	DEF
Mean	0.04	1.90	0.002	0.01	0.02	0.02	0.01	0.006	0.004
Standard deviation	1.15	0.73	0.62	0.68	0.91	0.47	0.42	0.002	0.003
Minimum	-17.44	0.68	-11.15	-5.03	-14.37	-2.97	-5.31	0.003	-0.005
Maximum	11.36	10.77	6.08	6.73	7.14	4.57	2.48	0.017	0.010
Panel B: Correlation coefficients									
Factor	$R_m - R_f$	σ_m	SMB	HML	MOM	RMW	CMA	$TERM$	DEF
$R_m - R_f$	1.00	0.03	-0.08	-0.29	-0.17	-0.29	-0.33	-0.01	0.00
σ_m		1.00	-0.02	0.03	-0.10	0.00	-0.02	-0.07	0.27
SMB			1.00	0.14	-0.09	-0.28	0.02	0.02	0.02
HML				1.00	-0.32	0.15	0.54	0.02	-0.01
MOM					1.00	0.09	0.05	-0.01	-0.05
RMW						1.00	0.25	-0.01	0.01
CMA							1.00	0.01	0.00
$TERM$								1.00	0.33
DEF									1.00

Table 2: Out-of-sample Fama-MacBeth cross-sectional tests for bank stock portfolios using different asset pricing models in the sample period January 1987 to December 2023. Using 25 size-sorted bank stock portfolios, this table contains results for the analysis period January 1988 to December 2023. Based on standard two-step Fama-MacBeth cross-sectional tests, we report the average of out-of-sample (one-month-ahead) estimated market prices of risk denoted $\hat{\lambda}_k$ for the k th factor. Associated t -statistics are shown in parentheses. The asset pricing models are: CAPM, Fama and French three-factor model (FF3), Carhart four-factor model (C4), Fama and French five-factor model (FF5), Fama and French six-factor model (FF6), and MVP model. Factors in these models are denoted as m (value-weighted CRSP excess return), σ_m (market return dispersion), SMB (size), HML (value), MOM (momentum), RMW (profitability), and CMA (capital investment).

Panel A: 25 size-sorted portfolios									
Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{\sigma_m}$	$\hat{\lambda}_{SMB}$	$\hat{\lambda}_{HML}$	$\hat{\lambda}_{MOM}$	$\hat{\lambda}_{RMW}$	$\hat{\lambda}_{CMA}$	R^2
CAPM	1.78 (8.40***)	-1.09 (-3.88***)							0.43
FF3	1.71 (7.97***)	-0.67 (-1.96**)		-0.09 (-0.37)	-0.21 (-0.72)				0.46
C4	1.53 (7.28***)	-0.65 (-1.90*)		0.01 (0.03)	-0.09 (-0.31)	-1.13 (-2.46**)			0.68
FF5	1.60 (7.72***)	-0.71 (-2.16**)		-0.08 (-0.34)	-0.07 (-0.24)		-0.42 (-1.83*)	-0.03 (-0.12)	0.62
FF6	1.49 (7.18***)	-0.73 (-2.18**)		0.04 (0.15)	0.00 (-0.01)	-1.26 (-2.64**)	-0.34 (-1.47)	0.09 (0.36)	0.78
MVP	1.26 (5.57***)	-0.72 (-2.28**)	0.49 (4.19***)						0.99
Panel B: 25 beta-sorted portfolios									
Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{\sigma_m}$	$\hat{\lambda}_{SMB}$	$\hat{\lambda}_{HML}$	$\hat{\lambda}_{MOM}$	$\hat{\lambda}_{RMW}$	$\hat{\lambda}_{CMA}$	R^2
CAPM	0.80 (4.49***)	0.09 (0.35)							0.13
FF3	0.78 (4.26***)	0.08 0.28		-0.15 -0.73	0.10 0.39				0.37
C4	0.78 (4.26***)	-0.08 (-0.26)		-0.17 (-0.79)	0.11 (0.44)	-0.13 (-0.32)			0.37
FF5	0.75 (3.93***)	0.10 (0.31)		-0.05 (-0.23)	0.04 (0.14)		0.06 (0.27)	-0.18 (-0.74)	0.38
FF6	0.74 (3.95***)	-0.06 (-0.19)		-0.05 (-0.23)	0.11 (0.40)	-0.30 (-0.72)	0.11 (0.46)	-0.15 (-0.63)	0.40
MVP	0.80 (4.19***)	0.09 (0.32)	-0.21 (-2.27**)						0.82

Table 2, continued

Panel C: 25 zeta-sorted portfolios									
Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{\sigma_m}$	$\hat{\lambda}_{SMB}$	$\hat{\lambda}_{HML}$	$\hat{\lambda}_{MOM}$	$\hat{\lambda}_{RMW}$	$\hat{\lambda}_{CMA}$	R^2
CAPM	0.48 (1.68*)	0.72 (1.58)							0.24
FF3	0.52 (1.88*)	0.80 (1.59)		-0.61 (-2.21**)	0.32 (0.89)				0.45
C4	0.47 (1.64)	1.02 (2.10**)		-0.62 (-2.36**)	0.23 (0.69)	0.15 (0.36)			0.43
FF5	0.60 (2.24**)	0.66 (1.44)		-0.55 (-1.98**)	0.32 (0.91)		0.21 (0.82)	0.25 (0.97)	0.56
FF6	0.58 (2.14**)	0.82 (1.71*)		-0.54 (-1.99**)	0.19 (0.58)	0.11 (0.26)	0.20 (0.85)	0.21 (0.87)	0.64
MVP	0.35 (1.22)	0.75 (1.69*)	0.22 (2.50**)						0.65
Panel D: 25 beta-zeta-sorted portfolios									
Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{\sigma_m}$	$\hat{\lambda}_{SMB}$	$\hat{\lambda}_{HML}$	$\hat{\lambda}_{MOM}$	$\hat{\lambda}_{RMW}$	$\hat{\lambda}_{CMA}$	R^2
CAPM	0.77 (4.41***)	0.12 (0.45)							0.10
FF3	0.89 (4.79***)	0.28 (0.91)		-0.41 (-1.66*)	-0.06 (-0.20)				0.14
C4	0.90 (4.79***)	0.35 (1.12)		-0.32 (-1.42)	0.01 (0.05)	-0.09 (-0.22)			0.36
FF5	0.96 (5.25***)	0.11 (0.37)		-0.37 (-1.45)	-0.02 (-0.06)		0.18 (0.71)	0.25 (0.94)	0.32
FF6	0.92 (5.03***)	0.15 (0.49)		-0.28 (-1.15)	0.12 (0.41)	-0.10 (-0.25)	0.30 (1.21)	0.29 (1.13)	0.44
MVP	0.89 (4.73***)	-0.03 (-0.10)	0.18 (1.94*)						0.74

Table 3: Out-of-sample Fama-MacBeth cross-sectional tests for bank stock portfolios using cross section factors in different asset pricing models in the sample period January 1987 to December 2023

Using 25 size-sorted bank stock portfolios, this table contains results for the analysis period January 1988 to December 2023. Based on standard two-step Fama-MacBeth cross-sectional tests, we report the average of out-of-sample (one-month-ahead) estimated market prices of risk denoted $\hat{\lambda}_k$ for the k th factor. Associated t -statistics are shown in parentheses. The asset pricing models are: CAPM, Fama and French three-factor model (FF3), Carhart four-factor model (C4), Fama and French five-factor model (FF5), and Fama and French six-factor model (FF6). For all factors except the market factor m (value-weighted CRSP excess return), cross section factors in these models are constructed as discussed in the text and denoted as *CSMB* (size), *CHML* (value), *CMOM* (momentum), *CRMW* (profitability), and *CCMA* (capital investment).

Panel A: 25 size-sorted portfolios								
Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{CSMB}$	$\hat{\lambda}_{CHML}$	$\hat{\lambda}_{CMOM}$	$\hat{\lambda}_{CRMW}$	$\hat{\lambda}_{CCMA}$	R^2
FF3	1.74 (8.14***)	-1.04 (-3.46***)	-0.27 (-0.16)	-3.64 (-1.46)				0.59
FF4	1.70 (8.31***)	-1.03 (-3.45***)	-1.03 (-0.61)	-3.20 (-1.31)	1.46 (0.35)			0.59
FF5	1.70 (8.27***)	-0.89 (-2.91***)	0.27 (0.16)	-2.67 (-1.06)		2.89 (1.10)	-3.36 (-1.13)	0.75
FF6	1.66 (8.35***)	-0.88 (-2.89***)	-0.11 (-0.07)	-2.88 (-1.14)	2.63 (0.61)	2.34 (0.89)	-2.42 (-0.83)	0.77
Panel B: 25 beta-sorted portfolios								
Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{CSMB}$	$\hat{\lambda}_{CHML}$	$\hat{\lambda}_{CMOM}$	$\hat{\lambda}_{CRMW}$	$\hat{\lambda}_{CCMA}$	R^2
FF3	0.82 (4.56***)	0.16 (0.58)	2.68 (1.81*)	0.33 (0.17)				0.46
FF4	0.92 (4.90***)	0.05 (0.19)	3.24 (2.20**)	0.16 (0.08)	1.23 (0.37)			0.48
FF5	0.83 (4.52***)	0.17 (0.59)	2.66 (1.65*)	0.06 (0.03)		-1.07 (-0.45)	-1.18 (-0.46)	0.57
FF6	0.88 (4.70***)	0.15 (0.51)	3.07 (1.94*)	-0.37 (-0.18)	0.32 (0.10)	-1.77 (-0.73)	-1.31 (-0.51)	0.61

Table 3, continued

Panel C: 25 zeta-sorted portfolios								
Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{CSMB}$	$\hat{\lambda}_{CHML}$	$\hat{\lambda}_{CMOM}$	$\hat{\lambda}_{CRMW}$	$\hat{\lambda}_{CCMA}$	R^2
FF3	0.58 (2.20**)	0.38 (0.87)	2.43 (1.56)	1.67 (0.79)				0.47
FF4	0.59 (2.28**)	0.45 (1.05)	1.86 (1.18)	0.94 (0.44)	-3.01 (-0.81)			0.50
FF5	0.43 (1.57)	0.59 (1.29)	2.32	0.29 (1.49)	(0.14)	4.99	0.55 (2.11)	0.47 (0.20)
FF6	0.45 (1.67*)	0.62 (1.38)	1.88 (1.21)	0.02 (0.01)	-1.57 (-0.42)	4.71 (2.00**)	0.83 (0.32)	0.49
Panel D: 25 beta-zeta-sorted portfolios								
Model	$\hat{\alpha}$	$\hat{\lambda}_m$	$\hat{\lambda}_{CSMB}$	$\hat{\lambda}_{CHML}$	$\hat{\lambda}_{CMOM}$	$\hat{\lambda}_{CRMW}$	$\hat{\lambda}_{CCMA}$	R^2
FF3	0.71 (4.11***)	0.18 (0.64)	0.37 (0.25)	0.91 (0.41)				0.15
FF4	0.76 (4.31***)	0.20 (0.68)	-0.63 (-0.39)	-0.15 (-0.07)	-1.97 (-0.50)			0.22
FF5	0.71 (4.22***)	0.08 (0.27)	0.08 (0.05)	1.77 (0.81)		-2.67 (-1.04)	-0.49 (-0.20)	0.37
FF6	0.79 (4.56***)	0.06 (0.21)	-0.81 (-0.46)	0.36 (0.16)	1.45 (0.37)	-3.26 (-1.23)	-1.23 (-0.49)	0.42

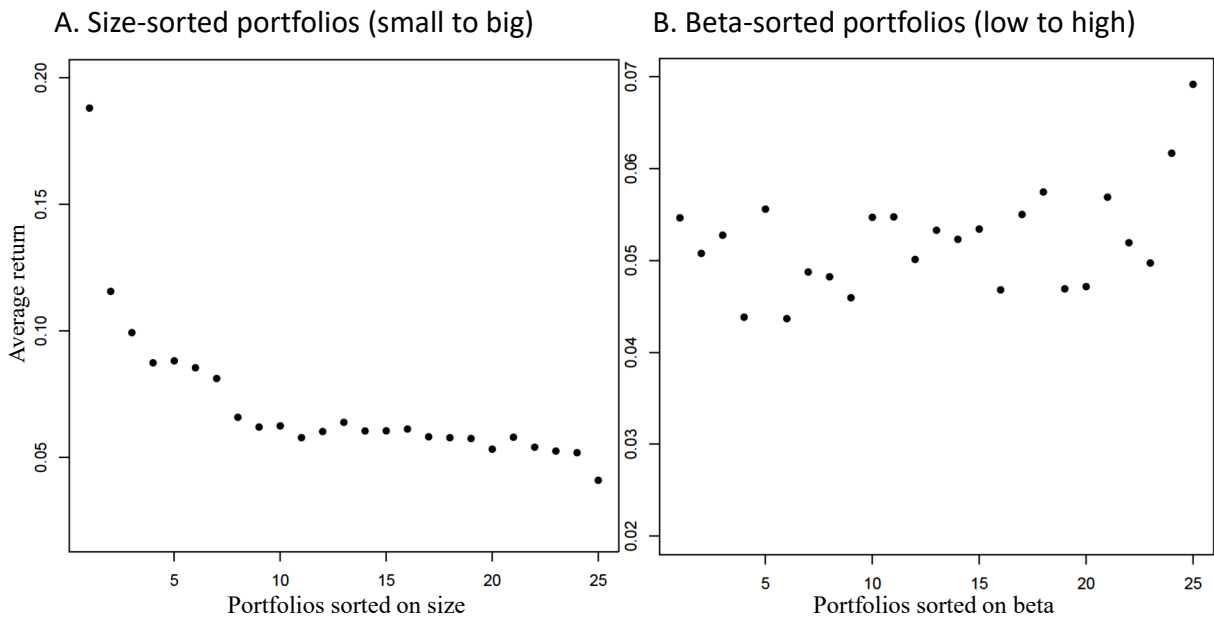


Figure 1: Mean daily returns of bank stocks sorted into 25 size portfolios from small to big (Panel A) and 25 beta portfolios from low to high (Panel B). The sample period is January 1987 to December 2023

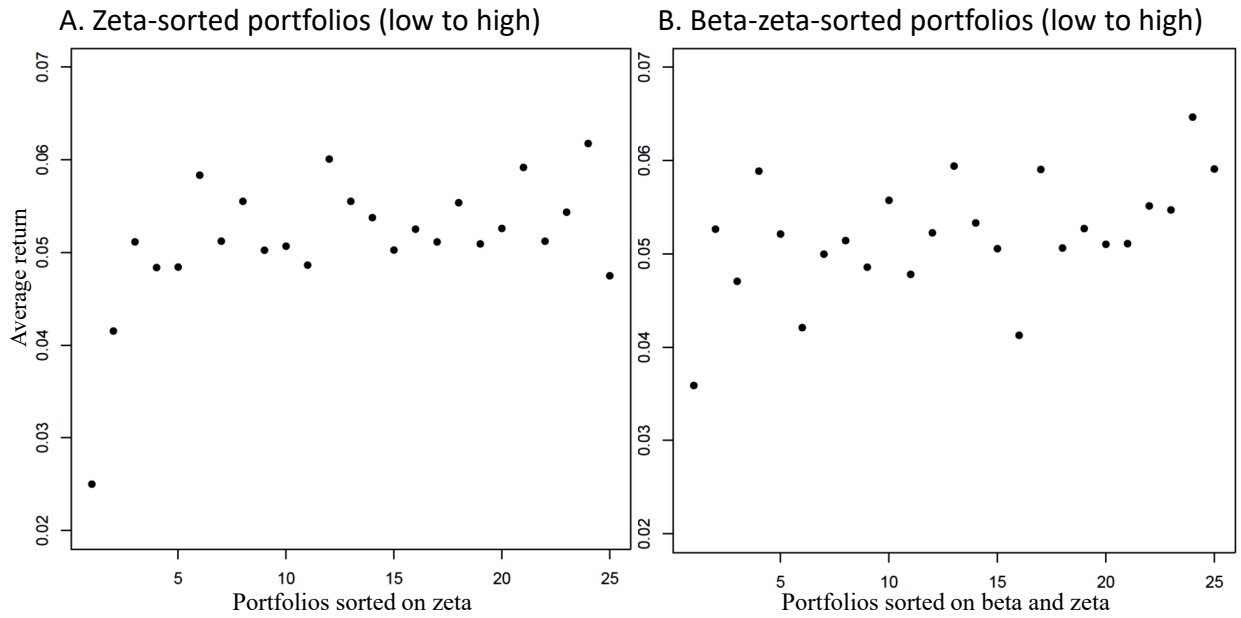


Figure 2: Mean daily returns of bank stocks sorted into 25 zeta portfolios from low to high (Panel A) and 25 beta-zeta portfolios from low to high (Panel B). The latter portfolios are formed from quintiles of beta and zeta portfolios, which are sorted on a beta quintile and then zeta quintiles within this beta quintile. The sample period is January 1987 to December 2023

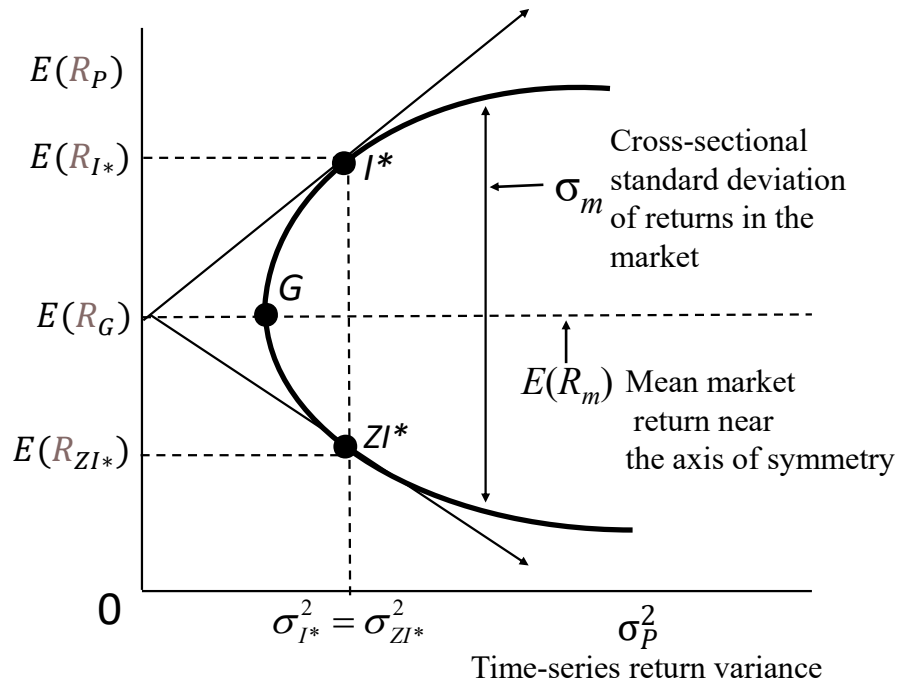


Figure 3: The Markowitz investment parabola, orthogonal portfolios I^* and ZI^* , and geometry of the ZCAPM.

Source: Adapted from Kolari, Liu, and Huang (2021, Figure 3.2, p. 59).

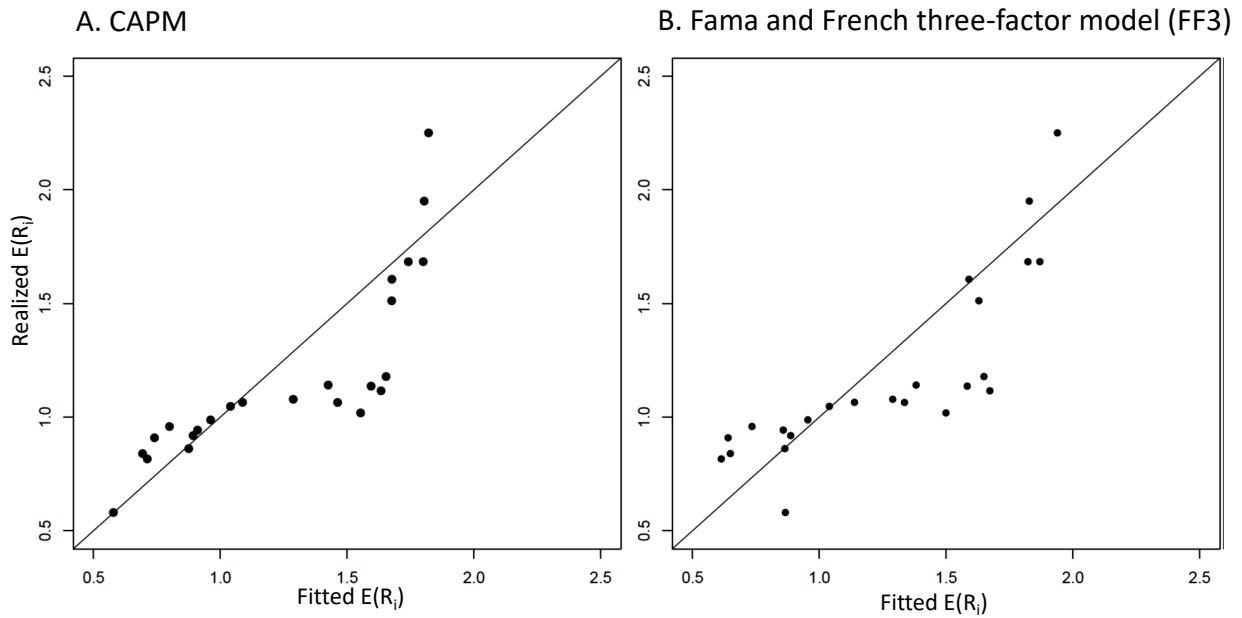


Figure 4: Out-of-sample cross-sectional CAPM (Panel A) and Fama and French three-factor model (Panel B) mispricing errors comparing average one-month-ahead realized excess returns in percent (Y-axis) to average one-month-ahead predicted (fitted) excess returns in percent (X-axis) for 25 size-sorted bank stock portfolios. The analysis period is January 1988 to December 2023

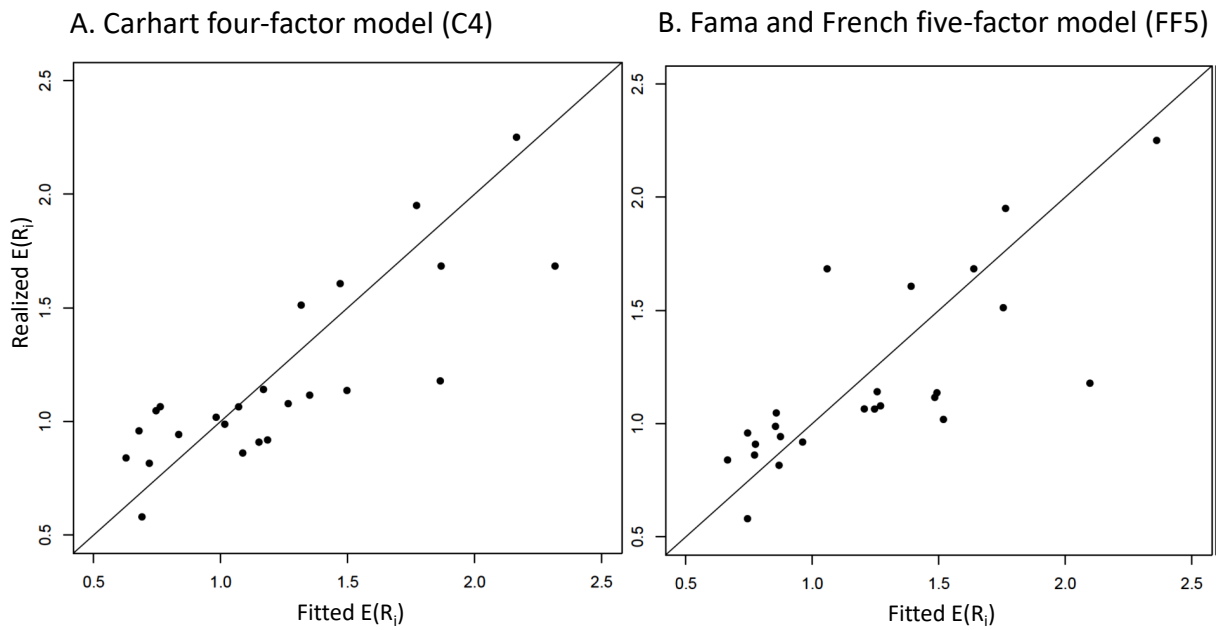


Figure 5: Out-of-sample cross-sectional Carhart four-factor model (Panel A) and Fama and French five-factor model (Panel B) mispricing errors comparing average one-month-ahead realized excess returns in percent (Y-axis) to average one-month-ahead predicted (fitted) excess returns in percent (X-axis) for 25 size-sorted bank stock portfolios. The analysis period is January 1988 to December 2023.

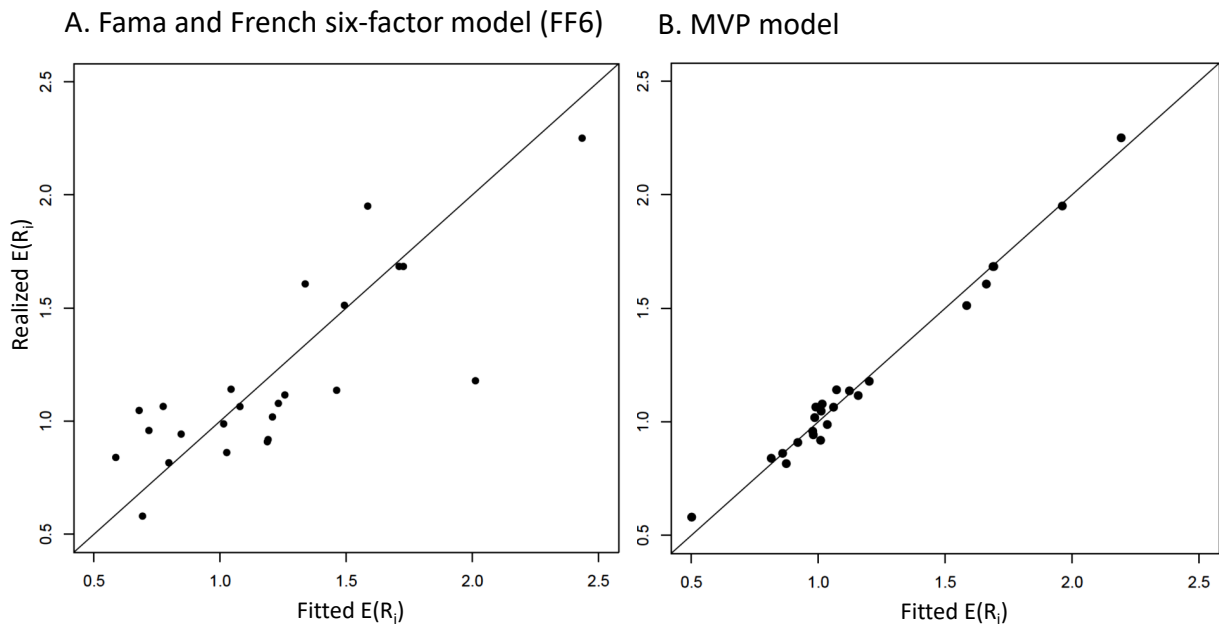


Figure 6: Out-of-sample cross-sectional Fama and French six-factor model (Panel A) and MVP model (Panel B) mispricing errors comparing average one-month-ahead realized excess returns in percent (Y-axis) to average one-month-ahead predicted (fitted) excess returns in percent (X-axis) for 25 size-sorted bank stock portfolios. The analysis period is January 1988 to December 2023

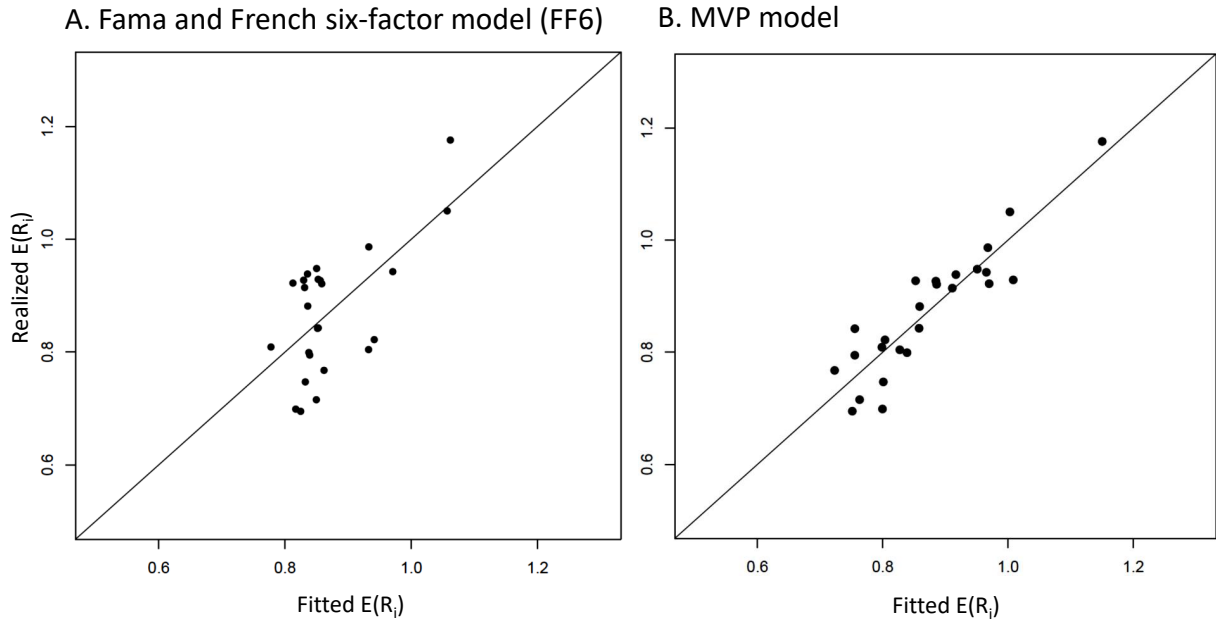


Figure 7: Out-of-sample cross-sectional Fama and French six-factor model (Panel A) and MVP model (Panel B) mispricing errors comparing average one-month-ahead realized excess returns in percent (Y-axis) to average one-month-ahead predicted (fitted) excess returns in percent (X-axis) for 25 beta-sorted bank stock portfolios. The analysis period is January 1988 to December 2023

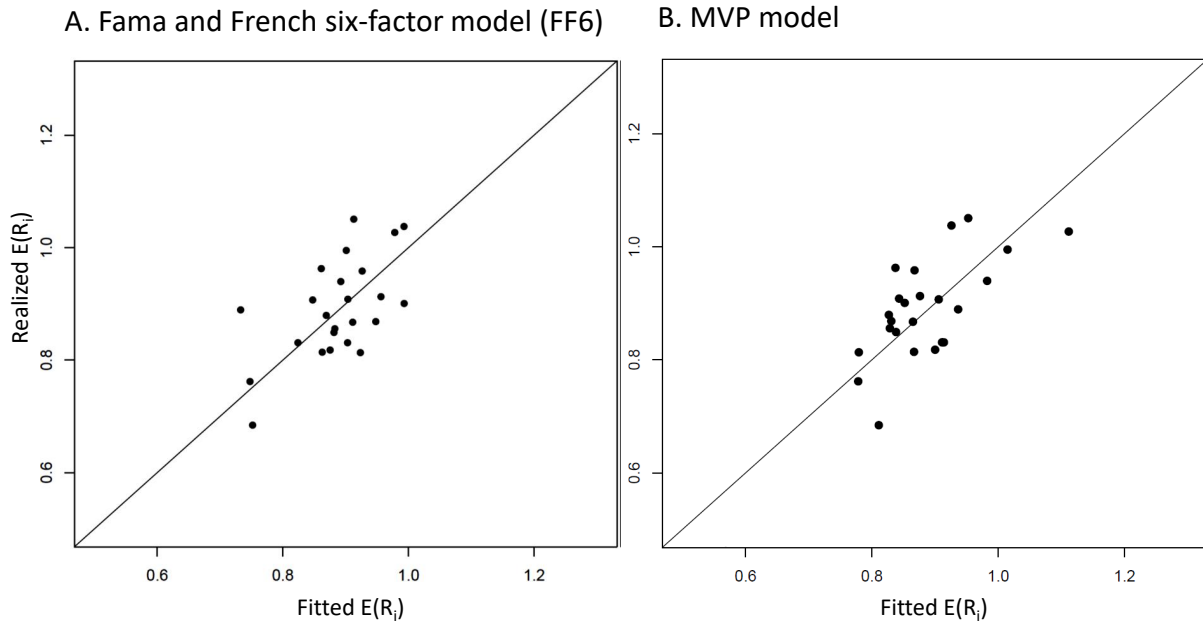


Figure 8: Out-of-sample cross-sectional Fama and French six-factor model (Panel A) and MVP model (Panel B) mispricing errors comparing average one-month-ahead realized excess returns in percent (Y-axis) to average one-month-ahead predicted (fitted) excess returns in percent (X-axis) for 25 zeta-sorted bank stock portfolios. The analysis period is January 1988 to December 2023

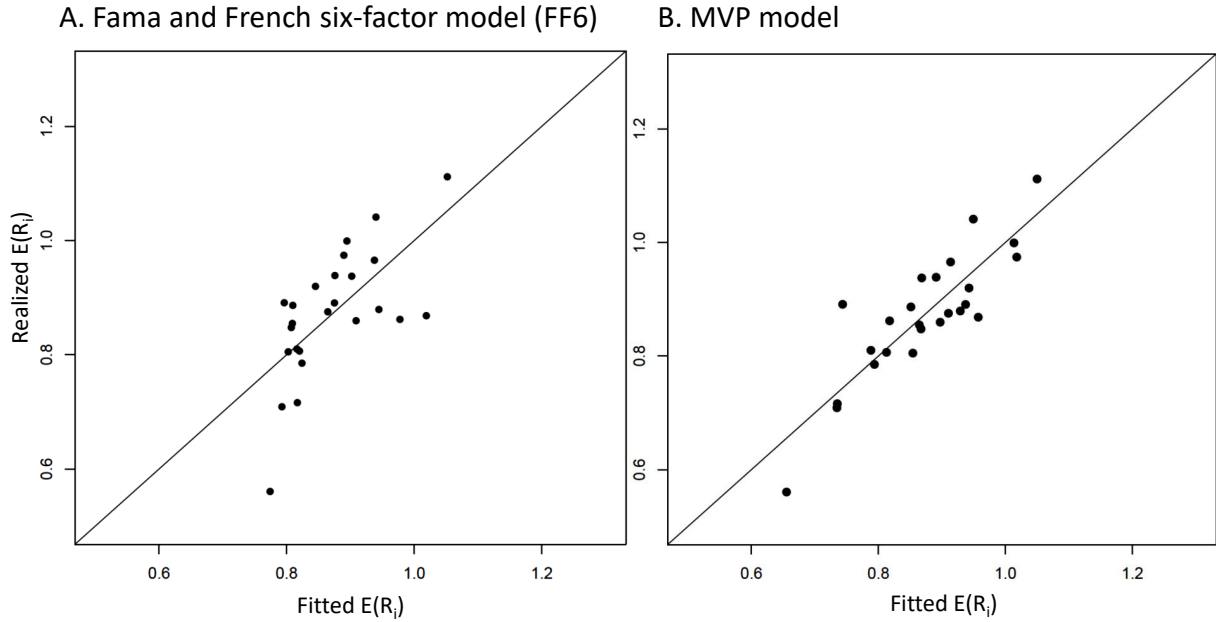


Figure 9: Out-of-sample cross-sectional Fama and French six-factor model (Panel A) and MVP model (Panel B) mispricing errors comparing average one-month-ahead realized excess returns in percent (Y-axis) to average one-month-ahead predicted (fitted) excess returns in percent (X-axis) for 25 beta-zeta-sorted bank stock portfolios. The analysis period is January 1988 to December 2023